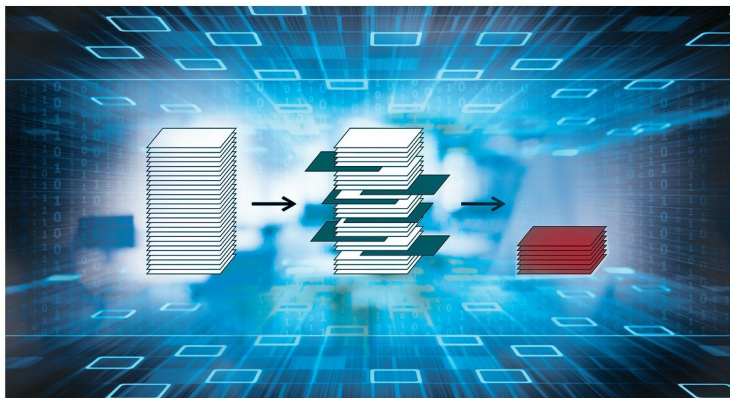


The AWK Programming Language: A Tool for Data Extraction

Introducing AWK, a programming language designed for text processing and typically used as a data extraction and reporting tool. This language is a standard feature of most UNIX-like operating systems.



AWK, one of the most prominent text-processing utilities on GNU/Linux, takes its name from the initials of its authors — Aho, Weinberger and Kernighan. It is an extremely versatile programming language that looks a little like C. It is a scripting language that processes data files, especially text files that are organised in rows and columns.

AWK really is a consistent tool with a few data types. Its portability and stability have made it very popular. It's a concise scripting language that can tackle a vast array of problems. It can teach the reader how to implement a database, a parser, an interpreter and a compiler for a small project-specific computer language.

If you are already aware of regex (regular expressions), it's quite easy to pick up the basics of AWK. This article will be useful for software developers, systems administrators, or any enthusiastic reader inclined to learn how to do text processing and data extraction in UNIX-like environments. Of course, one could use Perl or Python, but AWK makes it so much simpler with a concise single line command. Also, learning AWK is pretty low cost. You can learn the basics in less than an hour, so it doesn't require as much effort and time as learning any other programming/scripting language.

The original version of AWK was written in 1977 at AT&T Bell Laboratories. In 1985, a new version made the programming language more powerful, introducing user-defined functions, multiple input streams and computed regular expressions.

- AWK's first version came out in 1977 (old AWK)
- NAWK – this was an extended version that was released in 1985 (new AWK)
- MAWK – an extended version by Michael Brennan
- GAWK – GNU-AWK, which is faster and provides better error messages

Typical applications of AWK include generating reports, validating data, creating small databases, etc. AWK is very powerful and uses a simple programming language. It can solve complex text processing tasks with a few lines of code. Starting with an overview of AWK, its environment, and workflow, this article proceeds to explain its syntax, variables, operators, arrays, loops and functions.

AWK installation

Generally, AWK is available by default on most GNU/Linux distributions. We can use the `which` command to check whether